Autonomous fault prediction for rotating machines

Anouck Bruguière[1,2], Kais Hassan [1], Jean-Hugh Thomas [1], Amaury Graillat [2]

1. *Laboratoire d'Acoustique de l'Université du Mans (LAUM UMR CNRS 6613)*
2. *Oros Digital, Digigram Group*

anouck.bruguiere@oros.com

1 Context & Objective

Context

Unpredicted shutdowns are costly and AI traditional methods are limited:

- **Support Vector Machine (SVM)** characterized and limited by well-defined features [Liu18, Lei18]
- **Deep Learning (DL)** able to automatically extract features leading to an end-to-end diagnosis of the state of the system under study but with less explainable information

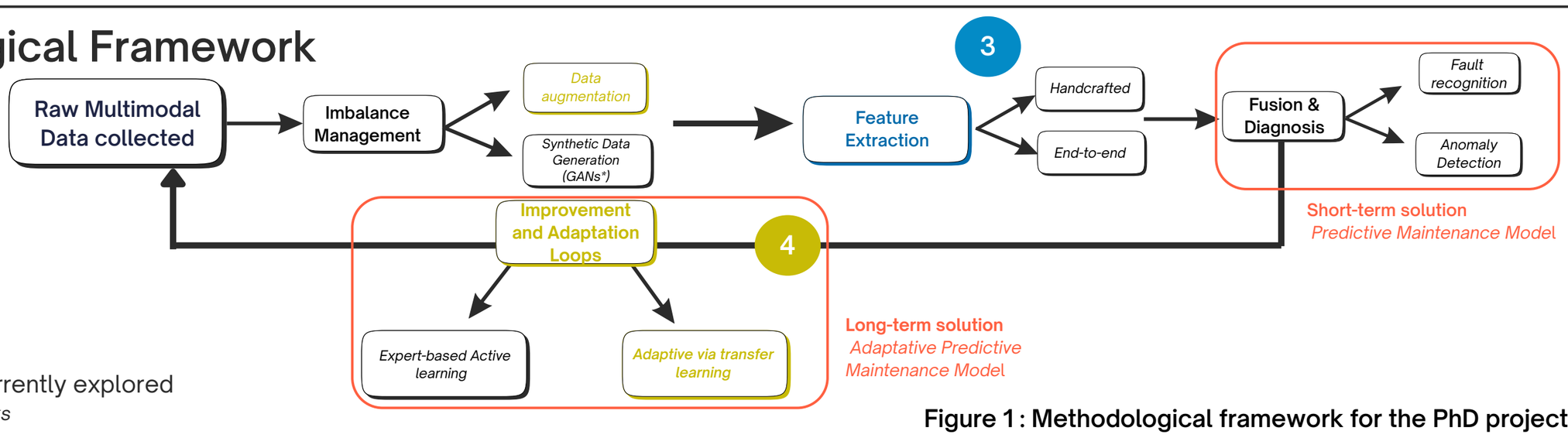
Contribution

Create a methodology for a model that can precisely identify a **fault's type**, **location**, and **severity**, of big rotating machines, leading to a more actionable and comprehensive diagnosis that relies on analyzing physical signals generated during operation, with vibration analysis being the most prominent method [Lei20].

OBJECTIVE

Prevent unplanned shutdowns in rotating machinery by developing an intelligent fault diagnosis framework, validated on a Francis Turbine.

2 Methodological Framework



3 & **4** : Solutions being currently explored

**Generative Adversarial Networks*

Figure 1 : Methodological framework for the PhD project

3 Feature-extraction Strategies

Which fault indicators from previous research work best across many different types of machines?

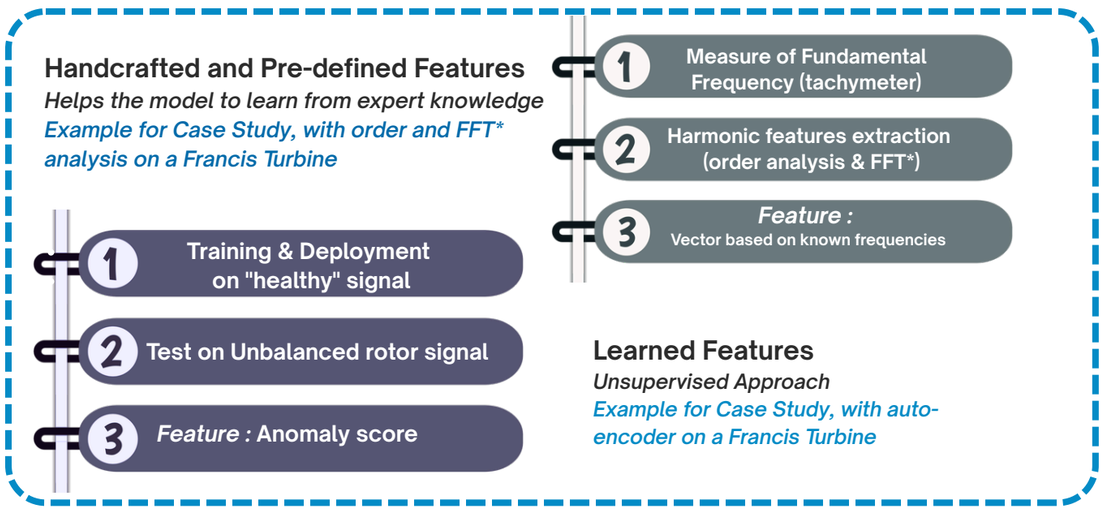


Figure 2 : Schematic of feature-extraction methods

4 Data issues

How can we gather enough data to cover all the possible faults a machine might have?

Key solutions

- **Data Scarcity:** *transfer learning (Cross-Condition or from a model), unsupervised & self-supervised learning for anomaly detection*
- **Trust & Explainability in DL** models:** *avoid a "black box" model, XAI*** techniques like SHAP, Multi-viz or LIME to visualize feature importance*
- **Robustness via Rigorous Uncertainty Quantification (UQ)**: *Hybrid models, physics-informed loss functions or dropout-based methods for uncertainty estimation (aleatoric uncertainty and epistemic uncertainty).* [Al17]

**Fast Fourier Transform*
***Deep Learning*
****Explainable Artificial Intelligence*

End-users are engineers and technicians, so these questions are essential if we are to win their trust.

5 Case study

Target Faults

- Unbalanced rotor
- Shaft Misalignment

Measured data

- **acceleration signal** of casing 1&2, and Plain bearing 1&2
- **Displacement** on axial
- **velocity** of the rotor

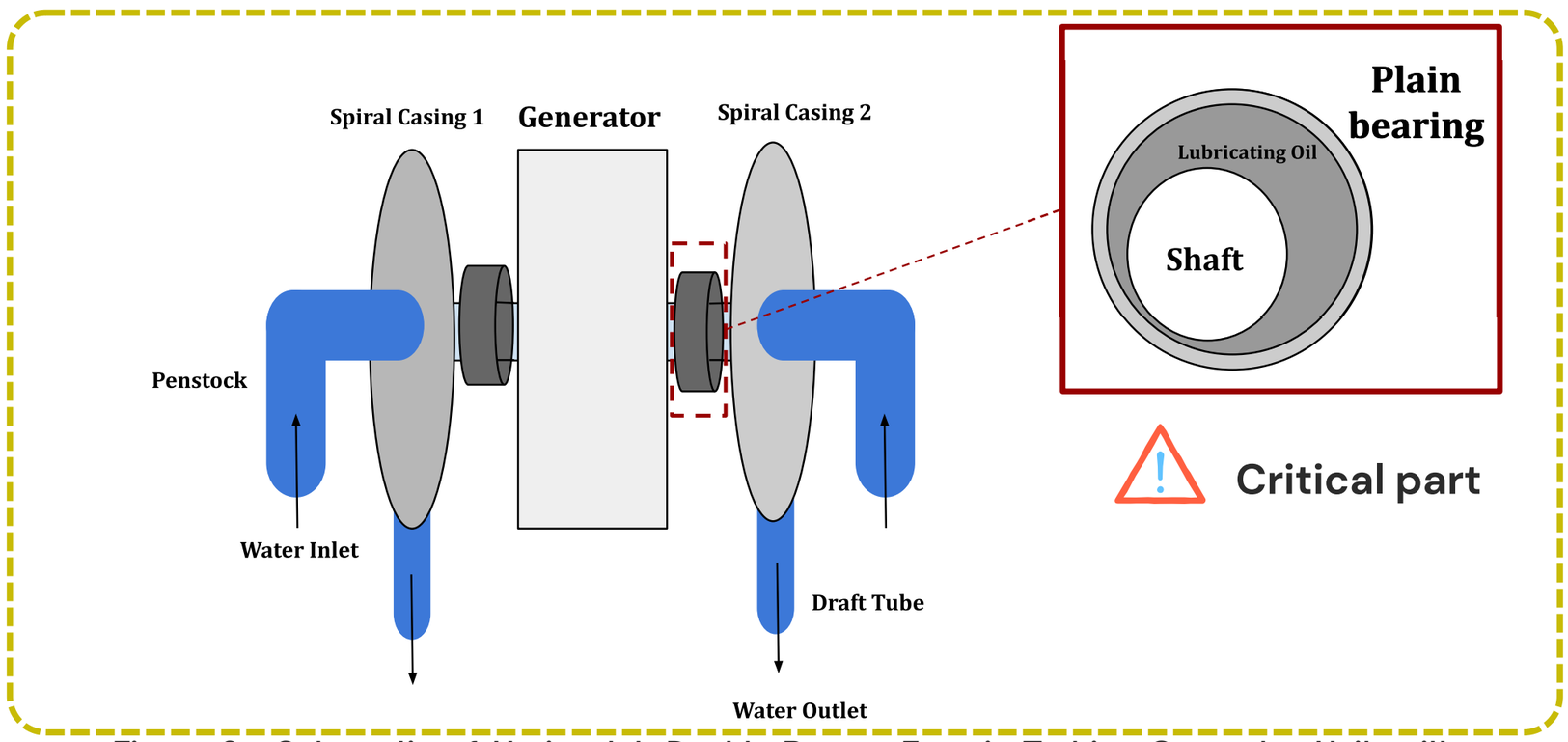


Figure 3 : Schematic of Horizontal, Double-Runner Francis Turbine Generator Unit, with cross-section showing the shaft floating on a fluid film

Related literature

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